

Mark Rober's CrunchLabs World's Largest Devil's Toothpaste

Time	Subtitle	Human Translation
0s	-[all] ...two, one. -[crashing]	-[all] ...two, one. -[crashing]
4s	[Mark] This is me reclaiming my title	[Mark] This is me reclaiming my title
6s	for the world's largest and tallest elephant toothpaste experiment.	for the world's largest and tallest elephant toothpaste experiment.
10s	And this is a different kind of reaction we discovered in all our experimentation	And this is a different kind of reaction we discovered in all our experimentation
14s	called devil's toothpaste,	called devil's toothpaste,
16s	which is insanely reactive	which is insanely reactive
18s	and super dangerous.	and super dangerous.
19s	I'll explain why in a minute, but for now, here's a hint.	I'll explain why in a minute, but for now, here's a hint.
23s	-[glass clattering] -[laughter]	-[glass clattering] -[laughter]
26s	Wow.	Wow.
27s	If you look closely in both shots, I'm not holding the trigger.	If you look closely in both shots, I'm not holding the trigger.
31s	This kid is.	This kid is.
32s	In fact, that kid was the whole reason we were there at all.	In fact, that kid was the whole reason we were there at all.
35s	This was a culmination of eight months of planning	This was a culmination of eight months of planning
37s	to throw him the most epic birthday party ever.	to throw him the most epic birthday party ever.
40s	And all of this started inadvertently from a single surprise phone call.	And all of this started inadvertently from a single surprise phone call.
49s	What's up, Fletcher?	What's up, Fletcher?
52s	-Oh my God. -[laughs]	-Oh my God. -[laughs]
59s	What's goin' on, buddy?	What's goin' on, buddy?
1:03	-You... What? -[laughs]	-You... What? -[laughs]
1:06	How are things, man?	How are things, man?
1:08	They just got better.	They just got better.
1:10	[Mark] This is Fletcher.	[Mark] This is Fletcher.
1:12	And this is also Fletcher, a year before this call.	And this is also Fletcher, a year before this call.
1:15	But unbeknownst to everyone at the time, also pictured here,	But unbeknownst to everyone at the time, also pictured here,
1:18	under all that amazing hair, was a fast-growing brain tumor.	under all that amazing hair, was a fast-growing brain tumor.
1:22	They realized something was wrong	They realized something was wrong

1:23	and discovered it was a dangerous form of brain cancer so rare,	and discovered it was a dangerous form of brain cancer so rare,
1:26	he was only the sixth kid ever known to have had it.	he was only the sixth kid ever known to have had it.
1:29	So his family uprooted their lives for a year	So his family uprooted their lives for a year
1:31	and moved almost 1,000 miles to be closer to St. Jude,	and moved almost 1,000 miles to be closer to St. Jude,
1:34	where they removed the tumor and would try and fix his brain.	where they removed the tumor and would try and fix his brain.
1:37	In fact, our call was the night before he was going back in	In fact, our call was the night before he was going back in
1:40	for another brutal week of chemo.	for another brutal week of chemo.
1:42	He's such a funny kid	He's such a funny kid
1:43	that I was so inspired by his hopeful attitude that, after talking for an hour,	that I was so inspired by his hopeful attitude that, after talking for an hour,
1:47	I secretly vowed that if he made it to 13 years old,	I secretly vowed that if he made it to 13 years old,
1:49	I would fly him and his family out	I would fly him and his family out
1:52	and surprise him with just the dopest birthday party ever.	and surprise him with just the dopest birthday party ever.
1:55	And not long after, I texted Fletcher this picture.	And not long after, I texted Fletcher this picture.
1:57	So he knew we had closely matching hairstyles,	So he knew we had closely matching hairstyles,
2:00	but what he didn't know is that plans were already underway	but what he didn't know is that plans were already underway
2:03	for him and I to take back the record	for him and I to take back the record
2:05	for the world's largest elephant toothpaste.	for the world's largest elephant toothpaste.
2:07	As a quick refresher, last summer, my buddy Science Bob and I	As a quick refresher, last summer, my buddy Science Bob and I
2:10	set the world record for elephant toothpaste	set the world record for elephant toothpaste
2:13	by filling up an empty swimming pool.	by filling up an empty swimming pool.
2:15	I wanted to be able to swim in the foam after,	I wanted to be able to swim in the foam after,
2:17	so we figured out how to use yeast as the catalyst,	so we figured out how to use yeast as the catalyst,
2:19	which is in fact safer, but it doesn't react very fast.	which is in fact safer, but it doesn't react very fast.
2:23	And our video seemed to inspire some folks because, two months later,	And our video seemed to inspire some folks because, two months later,
2:26	this happened in David Dobrik's backyard, where they didn't bother with yeast	this happened in David Dobrik's backyard, where they didn't bother with yeast

2:30	and just took the normal reaction and scaled it up.	and just took the normal reaction and scaled it up.
2:33	A few weeks later, these guys in Brazil did the largest experiment yet.	A few weeks later, these guys in Brazil did the largest experiment yet.
2:36	And they are the current Guinness World Record holders.	And they are the current Guinness World Record holders.
2:39	So even though Fletcher's birthday was eight months out,	So even though Fletcher's birthday was eight months out,
2:41	we started testing right away	we started testing right away
2:43	because I knew blasting a world-record amount of foam	because I knew blasting a world-record amount of foam
2:46	straight up into the sky	straight up into the sky
2:48	wasn't as easy as it might sound.	wasn't as easy as it might sound.
2:50	The part that concerned me most was picking the right nozzle size.	The part that concerned me most was picking the right nozzle size.
2:53	Here's what I mean.	Here's what I mean.
2:54	These three containers are identical except for the diameter of their outlets.	These three containers are identical except for the diameter of their outlets.
2:59	If that diameter's too large, it looks kind of cool	If that diameter's too large, it looks kind of cool
3:02	but it doesn't really go that high.	but it doesn't really go that high.
3:04	Too small, and it will go high	Too small, and it will go high
3:06	but it just looks like confetti and doesn't look that impressive.	but it just looks like confetti and doesn't look that impressive.
3:10	But for every container, there's a Goldilocks-size diameter opening	But for every container, there's a Goldilocks-size diameter opening
3:14	that gives you that cool laminar-looking flow	that gives you that cool laminar-looking flow
3:16	while still going pretty high.	while still going pretty high.
3:18	You might think, why don't you take a typical flask that works	You might think, why don't you take a typical flask that works
3:21	and scale the dimensions up?	and scale the dimensions up?
3:23	And we tried that on our first five-foot scale test...	And we tried that on our first five-foot scale test...
3:28	[Mark] And just as I feared, it went all confetti on us.	[Mark] And just as I feared, it went all confetti on us.
3:31	The problem is illustrated here.	The problem is illustrated here.
3:33	If you scale up and double all the dimensions of the block, you get this.	If you scale up and double all the dimensions of the block, you get this.
3:37	It's perfect. All these dimensions increased exactly by a factor of two.	It's perfect. All these dimensions increased exactly by a factor of two.
3:42	But wait a sec. The volume increased by a factor of eight.	But wait a sec. The volume increased by a factor of eight.
3:46	This is the square-cube law. Its effects are everywhere in nature,	This is the square-cube law. Its effects are everywhere in nature,

3:49	such as putting a hard limit on just how big land animals can be	such as putting a hard limit on just how big land animals can be
3:53	because how heavy you are is a function of your volume,	because how heavy you are is a function of your volume,
3:56	but the strength of your muscles and bones	but the strength of your muscles and bones
3:58	is a function of their cross-sectional area.	is a function of their cross-sectional area.
4:01	Since animals evolved to get heavier by the power of three	Since animals evolved to get heavier by the power of three
4:04	but stronger by only the power of two,	but stronger by only the power of two,
4:07	at some point, it's just too much.	at some point, it's just too much.
4:09	So on the next test, we increased the top diameter	So on the next test, we increased the top diameter
4:12	by a mere three and a half inches.	by a mere three and a half inches.
4:14	But the pendulum swung too far the other way	But the pendulum swung too far the other way
4:16	because while the column looked smooth, it didn't go that high.	because while the column looked smooth, it didn't go that high.
4:19	And so, for the third test, we let the pendulum swing back just a tad	And so, for the third test, we let the pendulum swing back just a tad
4:23	and shaved down that diameter by just over an inch	and shaved down that diameter by just over an inch
4:26	to try and dial in that Goldilocks outlet size.	to try and dial in that Goldilocks outlet size.
4:29	And sure enough, the column went really high and looked great.	And sure enough, the column went really high and looked great.
4:33	But this test uncovered another issue.	But this test uncovered another issue.
4:35	With such a high volume of foam,	With such a high volume of foam,
4:36	the pressure was creating some really high forces in the flask.	the pressure was creating some really high forces in the flask.
4:40	I mean, that's a lot of damage.	I mean, that's a lot of damage.
4:42	Which, quite frankly, left us with only one option.	Which, quite frankly, left us with only one option.
4:47	[Jon] Oh, yes!	[Jon] Oh, yes!
4:51	[Mark] With the power of Flex Tape, we scaled up even further	[Mark] With the power of Flex Tape, we scaled up even further
4:54	to a ten-foot test,	to a ten-foot test,
4:55	and did our best to interpolate that Goldilocks outlet size.	and did our best to interpolate that Goldilocks outlet size.
5:00	-[man] Oh my God! -[woman] Ooh!	-[man] Oh my God! -[woman] Ooh!
5:02	-[man] Oh my God! -[woman] Ooh!	-[man] Oh my God! -[woman] Ooh!
5:05	-[man] Oh my God! -[woman] Ooh!	-[man] Oh my God! -[woman] Ooh!
5:08	[Mark] Again, the outlet size worked great,	[Mark] Again, the outlet size worked great,
5:10	but our joint construction wasn't prepared for the further scaled-up forces.	but our joint construction wasn't prepared for the further scaled-up forces.

5:14	In fact, score a fat W for the Flex Tape,	In fact, score a fat W for the Flex Tape,
5:17	where you could see it singlehandedly	where you could see it singlehandedly
	held the thing together	held the thing together
5:20	even as the screws and wood busted	even as the screws and wood busted
	through.	through.
5:23	We were realizing we had to make the	We were realizing we had to make the
	structure	structure
5:25	so overengineered and indestructible	so overengineered and indestructible
5:28	so the only option for the foam to escape	so the only option for the foam to escape
	the high pressure	the high pressure
5:31	would be going out the top.	would be going out the top.
5:32	While that wasn't a perfect result, I felt	While that wasn't a perfect result, I felt
	we'd learned enough from our failures	we'd learned enough from our failures
5:36	to design it strong enough for the final 20-	to design it strong enough for the final 20-
	foot world record version.	foot world record version.
5:40	In the end, we ran over 150 tests,	In the end, we ran over 150 tests,
5:43	not only to determine the right flask	not only to determine the right flask
	dimensions,	dimensions,
5:46	but things like, does neck length have an	but things like, does neck length have an
	effect?	effect?
5:49	What happens if you vary the soap	What happens if you vary the soap
	amounts?	amounts?
5:51	What effect do different brands of food	What effect do different brands of food
	coloring have?	coloring have?
5:54	These tests were insightful, but perhaps	These tests were insightful, but perhaps
	the most interesting finding	the most interesting finding
5:58	was stumbling upon something I've	was stumbling upon something I've
	affectionately named	affectionately named
6:02	devil's toothpaste.	devil's toothpaste.
6:03	You might be thinking, "Devil's	You might be thinking, "Devil's
	toothpaste?"	toothpaste?"
6:05	"That seems a bit extreme." Allow me to	"That seems a bit extreme." Allow me to
	demonstrate.	demonstrate.
6:08	This is a box of the standard formula of	This is a box of the standard formula of
	hydrogen peroxide,	hydrogen peroxide,
6:11	sodium iodide and soap	sodium iodide and soap
6:13	pretty much everyone uses to make	pretty much everyone uses to make
	elephant toothpaste.	elephant toothpaste.
6:16	Now here's the same volume of reactants	Now here's the same volume of reactants
6:18	only this time using devil's toothpaste.	only this time using devil's toothpaste.
6:21	See if you can spot the difference in real	See if you can spot the difference in real
	time.	time.
6:30	Not only is it hyperreactive, but the	Not only is it hyperreactive, but the
	reaction is much hotter	reaction is much hotter
6:33	so you get these really cool bubble	so you get these really cool bubble
	formations.	formations.

6:36	Here's one more.	Here's one more.
6:37	This is normal elephant's toothpaste in an Erlenmeyer flask.	This is normal elephant's toothpaste in an Erlenmeyer flask.
6:40	No big deal. People have been doing this for years.	No big deal. People have been doing this for years.
6:43	But once again, here's the same volume of reactants,	But once again, here's the same volume of reactants,
6:45	now only with devil's toothpaste.	now only with devil's toothpaste.
6:51	Given how explosive this is, this may be irresponsible of me,	Given how explosive this is, this may be irresponsible of me,
6:55	but while we watch the replay, I'll tell you how to make it.	but while we watch the replay, I'll tell you how to make it.
6:58	You start with 26...	You start with 26...
7:00	-Once that mixes, you're... -[bleep]	-Once that mixes, you're... -[bleep]
7:03	-...Oprah Winfrey... -[bleep]	-...Oprah Winfrey... -[bleep]
7:05	...torch on the leftover rabbit fur, and you're good to go.	...torch on the leftover rabbit fur, and you're good to go.
7:08	By taking some rigorous safety precautions,	By taking some rigorous safety precautions,
7:10	we also intended to set the first and therefore only world record	we also intended to set the first and therefore only world record
7:15	for the world's largest devil's toothpaste experiment	for the world's largest devil's toothpaste experiment
7:18	by filling six 55-gallon drums	by filling six 55-gallon drums
7:20	with 400 times the amount that broke the flask here.	with 400 times the amount that broke the flask here.
7:23	So after 150 tests and eight months of work,	So after 150 tests and eight months of work,
7:26	feeling more confident about our build plans and chemistry,	feeling more confident about our build plans and chemistry,
7:29	we packed everything up	we packed everything up
7:31	and made our way to the small farming town of Gridley, California.	and made our way to the small farming town of Gridley, California.
7:34	We chose Gridley because our friend Dale graciously agreed to let us use his land,	We chose Gridley because our friend Dale graciously agreed to let us use his land,
7:39	which is full of hundreds of acres of walnut trees.	which is full of hundreds of acres of walnut trees.
7:42	No, don't even think about it, Phat Gus.	No, don't even think about it, Phat Gus.
7:45	He also had a great backyard we could use.	He also had a great backyard we could use.
7:48	With only four days till Fletcher was flying out for the party,	With only four days till Fletcher was flying out for the party,
7:51	we got to work on a 20-second build montage.	we got to work on a 20-second build montage.

8:14	[Mark] It took four consecutive 14-hour backbreaking days of labor in the hot sun	[Mark] It took four consecutive 14-hour backbreaking days of labor in the hot sun
8:19	because with so much time to plan the most epic party ever,	because with so much time to plan the most epic party ever,
8:22	I just kept adding more and more ideas to the list.	I just kept adding more and more ideas to the list.
8:25	So in addition to the indestructible 20-foot flask	So in addition to the indestructible 20-foot flask
8:27	and devil's toothpaste mechanism,	and devil's toothpaste mechanism,
8:29	we needed to make a methane bubble fountain,	we needed to make a methane bubble fountain,
8:31	a 3D bubble printer, a fire tornado,	a 3D bubble printer, a fire tornado,
8:33	a slip-n-slide into a pool	a slip-n-slide into a pool
8:35	and an epic bubble pit.	and an epic bubble pit.
8:37	Even though I brought my buddies,	Even though I brought my buddies,
8:38	it was cool to see this turn into a modern-day small-town USA barn-raising.	it was cool to see this turn into a modern-day small-town USA barn-raising.
8:42	If we needed extra hands or equipment,	If we needed extra hands or equipment,
8:44	ten neighbors were there instantly to help us out.	ten neighbors were there instantly to help us out.
8:47	The morning of the big day, there was last-minute scrambling,	The morning of the big day, there was last-minute scrambling,
8:50	but things were looking good.	but things were looking good.
8:51	I'm seriously so nervous right now.	I'm seriously so nervous right now.
8:53	I love pranks, and a surprise is the most wholesome type of prank,	I love pranks, and a surprise is the most wholesome type of prank,
8:58	so I haven't been able to sleep for days just thinking about this.	so I haven't been able to sleep for days just thinking about this.
9:01	Like, every last detail to how this will go down.	Like, every last detail to how this will go down.
9:04	With one hour left, we did one final test run on everything.	With one hour left, we did one final test run on everything.
9:07	-First, the devil's toothpaste. -Two, one...	-First, the devil's toothpaste. -Two, one...
9:11	Perfecto.	Perfecto.
9:12	And then the 20-foot flask.	And then the 20-foot flask.
9:15	For the real reaction, the peroxide would fill the whole bottom pool where Jon was,	For the real reaction, the peroxide would fill the whole bottom pool where Jon was,
9:19	and the troughs would dump over all the catalyst at once, like this.	and the troughs would dump over all the catalyst at once, like this.
9:22	Guinness only lets you choose one world record category per event.	Guinness only lets you choose one world record category per event.
9:26	We were going for the tallest,	We were going for the tallest,
9:28	but when we did the math to scale the flask	but when we did the math to scale the flask
9:30	to the arbitrary height of 20 feet,	to the arbitrary height of 20 feet,

9:32	and this is a true story,	and this is a true story,
9:33	it was exactly one gallon less than the folks in Brazil	it was exactly one gallon less than the folks in Brazil
9:37	who set the world record for the largest.	who set the world record for the largest.
9:39	So don't mind me while I set these two gallon containers right here.	So don't mind me while I set these two gallon containers right here.
9:43	As people started gathering for the surprise,	As people started gathering for the surprise,
9:45	I hit the hardened streets of Gridley, California,	I hit the hardened streets of Gridley, California,
9:48	to look for some kind of last-minute disguise.	to look for some kind of last-minute disguise.
9:54	After a bit of a warm-up,	After a bit of a warm-up,
9:55	'cause these are moves I can do, and that's definitely me in the costume,	'cause these are moves I can do, and that's definitely me in the costume,
9:59	the moment I had spent eight months planning for had finally arrived.	the moment I had spent eight months planning for had finally arrived.
10:04	The cover story for the kids	The cover story for the kids
10:06	is they were coming to tour a walnut farm of a distant relative.	is they were coming to tour a walnut farm of a distant relative.
10:10	-[all] Surprise! -[party poppers popping]	-[all] Surprise! -[party poppers popping]
10:13	-[cheering] -[woman] Whoo!	-[cheering] -[woman] Whoo!
10:16	[Mark] To provide a juxtaposition,	[Mark] To provide a juxtaposition,
10:18	I wanted him to feel as awkward as possible for this part.	I wanted him to feel as awkward as possible for this part.
10:23	[Mark] When we gathered for the picture, I snuck up behind him.	[Mark] When we gathered for the picture, I snuck up behind him.
10:26	-I took off my mask. -[all] Cheese!	-I took off my mask. -[all] Cheese!
10:27	[Mark] All right. Cool.	[Mark] All right. Cool.
10:30	What do you guys wanna do next?	What do you guys wanna do next?
10:33	-What should we-- -You're kidding.	-What should we-- -You're kidding.
10:34	-[laughter] -[Mark] Wha...	-[laughter] -[Mark] Wha...
10:37	[Mark] His sister Isla also had no idea any of this was gonna happen.	[Mark] His sister Isla also had no idea any of this was gonna happen.
10:44	[Mark] What up, buddy?	[Mark] What up, buddy?
10:47	How are you? Get in here!	How are you? Get in here!
10:51	-I'm the bull-- -Mom.	-I'm the bull-- -Mom.
10:53	[mom] Aw!	[mom] Aw!
10:55	-What's up, dude? -[laughter]	-What's up, dude? -[laughter]
10:59	When I'm not making videos, I'm the Gridley mascot.	When I'm not making videos, I'm the Gridley mascot.
11:02	You're the mascot?	You're the mascot?
11:04	To stretch out the surprise, I wanted him to think I was only there to say hi.	To stretch out the surprise, I wanted him to think I was only there to say hi.

11:08	I had Dale suggest I take the kids for a ride in his dune buggy,	I had Dale suggest I take the kids for a ride in his dune buggy,
11:12	to which I reluctantly agreed, 'cause, you know, I'm a busy guy.	to which I reluctantly agreed, 'cause, you know, I'm a busy guy.
11:15	I could do it. Like, 15 minutes.	I could do it. Like, 15 minutes.
11:17	-You guys wanna go for a ride? -Yeah.	-You guys wanna go for a ride? -Yeah.
11:19	Is that cool? Let's get the crew. Your sister.	Is that cool? Let's get the crew. Your sister.
11:21	I had about, like, a half an hour, so...	I had about, like, a half an hour, so...
11:24	-It's good to meet you in real life. -Yeah.	-It's good to meet you in real life. -Yeah.
11:27	Yeah, let's go!	Yeah, let's go!
11:29	Whoo!	Whoo!
11:32	Don't tell Mom!	Don't tell Mom!
11:34	-Not a word. -Not a word, okay!	-Not a word. -Not a word, okay!
11:37	Watch out for walnuts.	Watch out for walnuts.
11:39	-That's... Whoa! -Oh!	-That's... Whoa! -Oh!
11:41	A single walnut grows in one of these, which is crazy to me.	A single walnut grows in one of these, which is crazy to me.
11:45	All of those are a single walnut.	All of those are a single walnut.
11:53	As we pulled up, I stopped the dune buggy in a precisely predetermined location	As we pulled up, I stopped the dune buggy in a precisely predetermined location
11:57	where the view of the flask was obscured by the truck.	where the view of the flask was obscured by the truck.
12:00	-It's your birthday. I got you something. - [Fletcher] Tomorrow's my real birthday.	-It's your birthday. I got you something. - [Fletcher] Tomorrow's my real birthday.
12:04	Oh, yeah, I'm one day off.	Oh, yeah, I'm one day off.
12:08	-That is so cool. -So, Fletch...	-That is so cool. -So, Fletch...
12:10	I've gotta, uh... I've gotta come clean, buddy.	I've gotta, uh... I've gotta come clean, buddy.
12:13	You and I today are gonna set a Guinness World Record	You and I today are gonna set a Guinness World Record
12:17	for the world's tallest elephant toothpaste.	for the world's tallest elephant toothpaste.
12:19	-Wait, what? -Are you kidding me?	-Wait, what? -Are you kidding me?
12:21	You are currently in a YouTube video.	You are currently in a YouTube video.
12:24	[Isla] Crazy!	[Isla] Crazy!
12:25	I want you to wave to 20 million of your newest friends.	I want you to wave to 20 million of your newest friends.
12:28	Hello! [laughing]	Hello! [laughing]
12:30	We're gonna put the elephant toothpaste in there and just blast it in the air.	We're gonna put the elephant toothpaste in there and just blast it in the air.
12:35	-[mockingly] "Yeah, I got 15 minutes." - [laughs]	-[mockingly] "Yeah, I got 15 minutes." - [laughs]
12:39	I actually fully cleared my schedule.	I actually fully cleared my schedule.

12:42	-Oh my God. -[laughing]	-Oh my God. -[laughing]
12:44	[Mark] With the surprise fully and successfully executed,	[Mark] With the surprise fully and successfully executed,
12:47	my heart rate finally lowered	my heart rate finally lowered
12:48	and I gave him a preview of the cool stuff we had lined up for the party.	and I gave him a preview of the cool stuff we had lined up for the party.
12:52	Then we headed to the backyard where he saw the setup and his mom.	Then we headed to the backyard where he saw the setup and his mom.
12:56	You are devious!	You are devious!
12:59	At this point, it had sunk in there would be no walnut farm tour	At this point, it had sunk in there would be no walnut farm tour
13:02	and that her and I had been conspiring for months,	and that her and I had been conspiring for months,
13:07	For now, it was time to party.	For now, it was time to party.
13:37	♪ And everybody starts to move ♪	♪ And everybody starts to move ♪
13:40	♪ As soon as Pata Pata starts to play... ♪	♪ As soon as Pata Pata starts to play... ♪
13:43	Nice bubble party, Mark Rober.	Nice bubble party, Mark Rober.
13:48	♪ Whoo ♪	♪ Whoo ♪
13:51	-♪ Whoo, whoo, whoo ♪ -♪ Hit it ♪	-♪ Whoo, whoo, whoo ♪ -♪ Hit it ♪
13:55	♪ Whoo ♪	♪ Whoo ♪
13:57	-♪ Whoo, whoo, whoo ♪ -♪ Hit it ♪	-♪ Whoo, whoo, whoo ♪ -♪ Hit it ♪
14:19	[Mark] Before we busted out the flaming bubbles or bubble pit,	[Mark] Before we busted out the flaming bubbles or bubble pit,
14:22	we headed across the street to the world's first,	we headed across the street to the world's first,
14:25	and therefore largest, devil's toothpaste experiment.	and therefore largest, devil's toothpaste experiment.
14:29	Here we go, Fletcher!	Here we go, Fletcher!
14:32	-Three, two, one! -[crowd] ...two, one!	-Three, two, one! -[crowd] ...two, one!
14:34	-Pull! -[pop]	-Pull! -[pop]
14:36	-[whooshing] -[gasping]	-[whooshing] -[gasping]
14:37	[crowd] ...two, one, pull!	[crowd] ...two, one, pull!
14:39	-[gasping] -[Mark] Whoa!	-[gasping] -[Mark] Whoa!
14:43	-[gasping] -[Mark] Whoa!	-[gasping] -[Mark] Whoa!
14:47	-[gasping] -[Mark] Whoa!	-[gasping] -[Mark] Whoa!
14:49	[Mark] Whoa!	[Mark] Whoa!
14:52	Yeah!	Yeah!
14:55	[Mark] Oh my God!	[Mark] Oh my God!
14:57	[man] Whoo-hoo!	[man] Whoo-hoo!
14:58	Science Bob hooked us up with the slo-mo shot.	Science Bob hooked us up with the slo-mo shot.
15:01	Whoa!	Whoa!

15:27	[Mark] This shot was from a GoPro on rails.	[Mark] This shot was from a GoPro on rails.
15:29	We weren't fast enough with the pull.	We weren't fast enough with the pull.
15:31	And for some reason I found this WALL-E-level pathetically sad.	And for some reason I found this WALL-E-level pathetically sad.
15:35	You can see even in slo-mo the reaction happens so incredibly fast.	You can see even in slo-mo the reaction happens so incredibly fast.
15:46	[Mark] I apologize for the gratuitous amount of slo-mo,	[Mark] I apologize for the gratuitous amount of slo-mo,
15:49	but this has got to be the coolest thing I've ever filmed.	but this has got to be the coolest thing I've ever filmed.
16:08	And afterwards, it kept oozing this rainbow Play-Doh extrusion thing.	And afterwards, it kept oozing this rainbow Play-Doh extrusion thing.
16:25	It just looked like some kind of whimsical alien landscape.	It just looked like some kind of whimsical alien landscape.
16:30	As you can see, over time it just sort of deflates itself.	As you can see, over time it just sort of deflates itself.
16:33	After that, we went back, ate some dinner	After that, we went back, ate some dinner
16:35	and had some amazing, not at all creepy Army of Fletcher face cupcakes,	and had some amazing, not at all creepy Army of Fletcher face cupcakes,
16:39	while the crew started prepping for the world-record finale.	while the crew started prepping for the world-record finale.
16:42	In the vein of cool stuff I'd need at my birthday party if I was turning 13,	In the vein of cool stuff I'd need at my birthday party if I was turning 13,
16:47	I, of course, had to bust out the Elon Musk flamethrower	I, of course, had to bust out the Elon Musk flamethrower
16:49	to light fire bubbles.	to light fire bubbles.
16:51	And finally, back to the yard	And finally, back to the yard
16:52	to top things off with the most amazing bubble pit ever.	to top things off with the most amazing bubble pit ever.
16:55	It was like belly-flopping on a cloud.	It was like belly-flopping on a cloud.
17:08	We realized it was taking way longer to fill than we had calculated,	We realized it was taking way longer to fill than we had calculated,
17:12	so it was all hands on deck to pull this off before the sun went down.	so it was all hands on deck to pull this off before the sun went down.
17:15	Oh, and this is cool.	Oh, and this is cool.
17:17	The bubble pit was dual-purpose, because, once empty,	The bubble pit was dual-purpose, because, once empty,
17:20	it served as a bunker to give a completely safe front-row seat	it served as a bunker to give a completely safe front-row seat
17:23	to whatever was about to happen.	to whatever was about to happen.
17:25	With the flask close to being filled,	With the flask close to being filled,
17:27	Fletcher asked to take a ride in the Tesla 'cause he'd never been in one.	Fletcher asked to take a ride in the Tesla 'cause he'd never been in one.
17:31	-I know this song. -This song.	-I know this song. -This song.

17:33	-Okay. -This is in, like, every video.	-Okay. -This is in, like, every video.
17:35	-You guys are in a Mark Rober video. - Whoa!	-You guys are in a Mark Rober video. - Whoa!
17:38	Per the request, I gave them a quick ride.	Per the request, I gave them a quick ride.
17:42	[Fletcher] Oh my God!	[Fletcher] Oh my God!
17:45	-[boy] Can this-- [Fletcher] Whoa!	-[boy] Can this-- [Fletcher] Whoa!
17:48	-Oh my-- -Look out!	-Oh my-- -Look out!
17:49	Autopilot.	Autopilot.
17:51	Oh my God!	Oh my God!
17:53	The weird part is I can sit and have a conversation with you,	The weird part is I can sit and have a conversation with you,
17:56	and this feels very freaky.	and this feels very freaky.
17:58	But you trust... [yells]	But you trust... [yells]
17:59	Remember our story, all right?	Remember our story, all right?
18:00	-Yeah. -Ten miles an hour, tops.	-Yeah. -Ten miles an hour, tops.
18:02	[Mark] With the measurement drone in the air, our timing was perfect.	[Mark] With the measurement drone in the air, our timing was perfect.
18:05	We made it back as they were sealing the door for the big moment.	We made it back as they were sealing the door for the big moment.
18:09	-I'm handing this to you officially. -[all] Ooh!	-I'm handing this to you officially. -[all] Ooh!
18:12	To think, you should have been touring a walnut farm right now.	To think, you should have been touring a walnut farm right now.
18:15	After eight months of planning and so much work by so many people,	After eight months of planning and so much work by so many people,
18:19	the coolest birthday present I've ever given anybody	the coolest birthday present I've ever given anybody
18:22	came down to this moment.	came down to this moment.
18:24	Here we go. Three, two, one.	Here we go. Three, two, one.
19:06	It worked!	It worked!
19:08	It went through the bottom!	It went through the bottom!
19:10	That was so cool!	That was so cool!
19:11	It came out the bottom. It lifted it up!	It came out the bottom. It lifted it up!
19:15	Look at my truck!	Look at my truck!
19:16	-Let's go. -[Isla] Oh my God!	-Let's go. -[Isla] Oh my God!
19:19	What'd you think?	What'd you think?
19:20	-So cool! -[boy] Insane.	-So cool! -[boy] Insane.
19:21	-[Fletcher] Crazy! -[Isla] It exploded!	-[Fletcher] Crazy! -[Isla] It exploded!
19:23	Look at the truck! It's, like, buried next to it.	Look at the truck! It's, like, buried next to it.
19:33	-[gasping] -[Mark] Watch this.	-[gasping] -[Mark] Watch this.
19:35	Boom!	Boom!

20:00	[Mark] I'd like to point out how great that column looks,	[Mark] I'd like to point out how great that column looks,
20:03	with a perfectly-selected outlet diameter.	with a perfectly-selected outlet diameter.
20:06	That is, until things got chaotic	That is, until things got chaotic
20:08	once the pressure changed drastically from the big rupture.	once the pressure changed drastically from the big rupture.
20:23	We literally made a rocket ship.	We literally made a rocket ship.
20:25	Fundamentally, this is how rockets propel themselves.	Fundamentally, this is how rockets propel themselves.
20:28	The actual failure mode wasn't a floor strength issue	The actual failure mode wasn't a floor strength issue
20:31	as much as it was a stiffness issue.	as much as it was a stiffness issue.
20:32	Our structure weighed over a ton.	Our structure weighed over a ton.
20:34	I didn't consider that the reaction could lift the whole thing	I didn't consider that the reaction could lift the whole thing
20:37	five feet into the freakin' air.	five feet into the freakin' air.
20:39	At NASA, that's what we refer to as a critical oversight.	At NASA, that's what we refer to as a critical oversight.
20:42	But this result was way more spectacular, plus it leaves room for improvement.	But this result was way more spectacular, plus it leaves room for improvement.
20:47	I should also note, even with all that,	I should also note, even with all that,
20:49	we still crushed the previous world record by more than double.	we still crushed the previous world record by more than double.
20:52	And so, as the sun went down,	And so, as the sun went down,
20:54	Science Bob lit up the fire tornado as a fitting end to the party.	Science Bob lit up the fire tornado as a fitting end to the party.
21:00	I also told Fletcher there was one more thing.	I also told Fletcher there was one more thing.
21:03	With help from Dale,	With help from Dale,
21:04	he could use some heavy equipment to destroy our beautiful creation.	he could use some heavy equipment to destroy our beautiful creation.
21:12	-[man 1] Whoa! -[man 2] Whoo-hoo-hoo!	-[man 1] Whoa! -[man 2] Whoo-hoo-hoo!
21:14	[Mark] And so with that, the party was all over.	[Mark] And so with that, the party was all over.
21:18	-That was epic. I'll never forget this. - [mom] Thank you.	-That was epic. I'll never forget this. - [mom] Thank you.
21:21	[Mark] As they drove away, I thought about everything they'd been through	[Mark] As they drove away, I thought about everything they'd been through
21:25	and all the people that came together from all across the country	and all the people that came together from all across the country
21:29	to donate their time and energy for a kid they'd never even met.	to donate their time and energy for a kid they'd never even met.
21:33	And I might not vote the exact same as all of them,	And I might not vote the exact same as all of them,

21:36	but for, like, five inspiring days, none of that stuff mattered.	but for, like, five inspiring days, none of that stuff mattered.
21:40	Whenever I find myself short on hope, I find it helps to put my phone down	Whenever I find myself short on hope, I find it helps to put my phone down
21:44	and choose to see the good parts in my fellow humans	and choose to see the good parts in my fellow humans
21:47	and be one of the good parts for my fellow humans to see.	and be one of the good parts for my fellow humans to see.
21:51	And so, to Fletcher, thank you for being such a good example	And so, to Fletcher, thank you for being such a good example
21:55	of holding onto hope amidst really lame circumstances.	of holding onto hope amidst really lame circumstances.
21:58	Happy birthday, buddy. And congrats on your new world record.	Happy birthday, buddy. And congrats on your new world record.